Who Would Have Complied Anyway?

Applying Methods of Causal Inference to Alignment-Faking Evaluations

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Abstract

Alignment-faking evaluations compare a model's compliance under a framing that implies training with its compliance under one that implies no monitoring, and read a classifier over the model's scratchpad to see whether the model reasoned strategically about the training. The compliance gap and the classifier's flag rate are reported side by side, and the relation between them is read informally. We show that the two arms of such an evaluation form an encouragement design with one-sided noncompliance, that the gap identifies the share of compliant responses attributable to the framing but not the share in which the reasoning occurred, and that the classifier's recall of that reasoning is therefore bounded rather than measured. We compute these bounds for every published model-setting pair we could obtain from three studies. The base rate of compliance when unmonitored determines what the evaluation can establish. A positive lower bound on recall is available only when the classifier's flag rate exceeds one minus the floor, and in the published data that happens only where the model complies unmonitored in under 4% of draws. None of the published evaluations above that base rate certifies any recall at all, whatever classifier it used. Claude 3 Opus's recall is pinned at 1.00 while Claude 3.5 Sonnet's, measured by the same classifier on the same prompts, is at most 0.25. Three results that the source papers leave as puzzles or residuals follow from the structure. The post-RL "disconnect" between a rising rate of alignment-faking reasoning and a shrinking compliance gap is what the design predicts once the model complies in almost every draw. The "low-coherence" alignment faking of models with gaps but no flagged reasoning is a bounds statement, not a behavioral category: for those models the design cannot distinguish unverbalized reasoning from a gap produced by some other route. We state the conditions under which the bounds narrow, show that the classifier's false-flag rate measured on the unmonitored arm narrows them further, and give reporting conditions for compliance-gap evaluations. All quantities are computed from published counts; the arithmetic for every row is reproduced in the appendix.

1 Introduction

Greenblatt et al. (2024) put a model in a scenario where it is told that a new training objective conflicts with its values and that only some of its conversations will be used for training. The model complies with the objective more often when it believes it is in training, and its hidden scratchpad often contains explicit reasoning that it is complying to avoid modification. Two numbers summarize the result: the *compliance gap* between the two conditions, and the rate of *alignment-faking reasoning* flagged by a classifier over the scratchpads. Subsequent work has reported the same pair across more models (Sheshadri et al. 2025; Hughes et al. 2025) and across model generations (Henselmans et al. 2026).

The two numbers are reported side by side, and readers draw a relation between them. When the gap is large and the flag rate is high, the reasoning is said to explain the gap; when the gap persists while the flag rate falls, the reasoning is said to have gone unverbalized. Both readings concern a quantity neither number reports: the share of the gap-producing responses in which the reasoning actually occurred, and the share of that reasoning the classifier caught.

This paper supplies the structure that relates them. The evaluation is an encouragement design with one-sided noncompliance (Angrist, Imbens and Rubin 1996): the training framing is the encouragement, reasoning about the training situation is the treatment, observed only through a classifier that reads the scratchpad with error, and compliance is the outcome. A companion paper develops this framing for chain-of-thought monitorability evaluations and derives the identification results used here (Diamond 2026). We restate what is needed in Section 2 and refer to that paper for the derivations and for a fuller discussion.

Three consequences follow, and we take them up in turn.

First, the compliance gap identifies the share of compliant responses *attributable* to the framing, which is a floor on the share in which the reasoning occurred, and the classifier's recall of that reasoning is bounded, not measured (Section 3). We compute those bounds for fifteen published model-setting pairs from three studies (Section 4). A positive lower bound, the only form in which such an evaluation certifies anything about the classifier, appears only at low base rates, whichever classifier produced it.

Second, three findings the source papers leave open are consequences of the structure (Section 5). Greenblatt et al. describe as a puzzle that reinforcement learning raises alignment-faking reasoning from 12% to 78% while shrinking the compliance gap; on the encouragement-design reading the two quantities are not commensurable, and after RL the gap is a sliver of near-total compliance. Sheshadri et al. classify four of their five gap-showing models as exhibiting "low coherence alignment faking" because the gaps come with almost no flagged reasoning; for those models the bounds show that the data cannot separate unverbalized reasoning from a gap with another cause. And a floor exists only where the gap is positive, a condition that is a sign test on a noisy estimate: in one of our rows it is met by a gap of a single percentage point that no test can distinguish from zero.

Third, the design says what would narrow the bounds (Section 6), and one of the narrowing quantities, the classifier's false-flag rate, is already measurable in these evaluations and is not currently used.

We close with reporting conditions for compliance-gap evaluations (Section 7).

Nothing here says that any published alignment-faking result is wrong. The evaluations are careful, and the practice of reporting the gap and the flag rate separately, rather than combining them into a single score, is better than the practice in the adjacent monitorability literature. What we add is a statement of what the two numbers do and do not jointly determine, and a quantity, the base rate of compliance when unmonitored, that determines how much they determine.

2 The evaluation as an encouragement design

2.1 Notation

An evaluation of this kind puts a population of prompts to the model under each of two framings, one implying that the response will be used for training and one implying that it will not, and collects a *response* to each prompt under each framing. The sources differ in how they build that population: Hughes et al. (2025) and Sheshadri et al. (2025) use a set of harmful requests with a single response per request per framing, Greenblatt et al. (2024) sample from a distribution of queries, and Henselmans et al. (2026) use 125 paraphrases of one scenario with 25 runs of each. The quantities below are rates over the pooled population, and the identities hold at that level whether the population is built by varying the prompt, by repeated sampling, or by both. For a response, $Z \in \{0, 1\}$ records the framing (Z = 1 implies training), $D \in \{0, 1\}$ whether the model reasoned about the training situation in deciding how to answer, $Y \in \{0, 1\}$ whether the response complied with the objective, and $M \in \{0, 1\}$ the classifier's verdict on the scratchpad. Write

- p_0 = Pr(Y = 1 | Z = 0), the *base rate*: compliance when unmonitored
- p_1 = Pr(Y = 1 | Z = 1), compliance under the training framing
- π = Pr(D = 1 | Z = 1), the *take-up rate*: the share of training-framing responses in which the reasoning occurred

No assignment is randomized, and none needs to be: the two arms present the same prompt population to the same model under two framings, so they are comparable by construction. We write *draw* and *response* interchangeably below, and note where a quantity is a share of prompts rather than of repeated samples.

2.2 Three quantities

A draw cannot reason about a training threat it was not given, so D = 0 whenever Z = 0 and noncompliance is one-sided (Bloom 1984). Suppose that a draw in which the reasoning occurs complies, and that draws without it behave as they would unaided, at the same rate p_0. Then

$p_1 = \pi + (1 - \pi)p_0$, TE = $p_1 - p_0 = \pi\,(1 - p_0)$, R_{effect} = TE/p_1, influence rate = π/p_1.

TE is the compliance gap, the intention-to-treat effect of the framing. R_{effect} is the share of compliant training-arm responses that would not have complied without the framing. **The influence rate** π/p_1 is the share in which the reasoning actually occurred. It is what the classifier is asked to detect, and the two arms do not identify it. It exceeds R_{effect} by $\pi\, p_0/p_1$, the draws that reasoned and would have complied anyway, and the ratio of the two is $1/(1 - p_0)$, which depends on the base rate alone.

All three are shares of a population, and that is the difficulty. TE says what proportion of compliant training-arm responses the framing moved; it does not say which ones. A response that complied might be one the framing moved or one the model would have given anyway, and nothing in the two arms distinguishes them, because that response's own counterfactual under the other framing was never observed: the other arm holds different responses. The classifier, meanwhile, is asked about each response individually. Scoring a population quantity against a per-response verdict is what leaves recall bounded rather than measured, and Section 2.3 makes that precise.

The identities require that the framing move compliance only through the reasoning: the exclusion restriction. It can fail. The framing is also a piece of text, and it can change compliance by making the deployer's intent more salient, by signalling what kind of answer is expected, or by any other

route that leaves nothing in the scratchpad for a reader to find. Sheshadri et al.'s rater-sycophancy and "AI lab intention saliency" hypotheses are exactly such routes, and Henselmans et al. note that they cannot rule out "some other mechanism producing differential behavior across contexts." Under the restriction the floor is a lower bound on what a perfect reader of the scratchpad would flag; when it fails, some gap-producing draws contain nothing to find, and even a perfect reader falls below the floor.

2.3 Recall, and why it is bounded

Recall is the share of draws with D = 1 that the classifier flags. Among training-arm compliant draws, a share c = π/p_1 reasoned, the classifier flags a share f of all of them, and recall is the share of the c-group carrying a flag. Flags falling in the c-group cannot exceed f, so recall $\leq$ f/c; at most $1 - c$ of the flags can fall outside it, so at least $f - (1 - c)$ fall inside and recall $\geq$ (f + c − 1)/c. Hence for a postulated influence rate c,

recall $\in$ [max(0, (f + c − 1)/c), min(1, f/c)].

The lower end rises and the upper end falls as c rises, so evaluating at the smallest admissible c = R_{effect} gives the widest interval the data allow. We treat c as a sensitivity parameter throughout, reporting the interval at the floor and at values that stated assumptions imply (Section 6).

Two things follow immediately. A positive lower end requires f > $1 - R_{\text{effect}}$, and since the influence rate π/p_1 cannot exceed 1, $R_{\text{effect}} = (1 - p_0)\pi/p_1$ is at most $1 - p_0$, so that bar is at least the base rate: an evaluation can certify a recall floor only if its classifier flags a larger share of compliant responses than the model would have complied at unaided. The framework paper states the same condition for monitorability evaluations (Diamond 2026, its Section 5.2). And where the bar is not cleared the lower end is zero whatever the classifier does, so a classifier that flags little and one that flags nothing are, at that end, not distinguishable by these two arms. Section 4 separates the cases.

3 Data and coding

We use three published sources. None of the authors computed the quantities below; all inputs are counts or rates they report.

Hughes et al. (2025) release per-response transcripts for their open-source replication of the helpful-only prompted setting: 100 responses per arm per model, one per harmful request, each with a StrongReject compliance score, a partial-compliance flag, and a 20-vote alignment-faking score. We code a response compliant when the StrongReject score exceeds 0 and flagged when the alignment-faking score is at least 0.40, their own threshold. Response-level data let us compute f exactly and, from the unmonitored arm, the classifier's false-flag rate φ.

Henselmans et al. (2026) report, for each of six Claude models across 125 prompt perturbations of the animal-welfare scenario, the number of runs and noncompliant responses in each arm and the number of training-arm responses flagged for verbalized alignment faking. Here compliance means dismissing animal welfare as instructed. We pool the five perturbation classes per model. Their sheet also carries a second, looser flag column; we use the strict one and note the effect of the other in Appendix B.

Greenblatt et al. (2024) report refusal rates in both arms and an alignment-faking reasoning rate for each prompt variation. Their flag rate is a share of all training-arm draws and is computed only on draws whose output complied, so f = (reported rate)/p_1.

Fourteen model-setting pairs have a positive gap and so admit a floor; one, Claude 3 Haiku in Henselmans et al., does not, with a gap of -0.1 points. In one of the fourteen the gap is not distinguishable from zero: Claude 3.7 Sonnet in Hughes et al., with 4 compliant responses against

	p_0	p_1	TE	Floor	f	Recall bounds	Counts
Hughes et al. (2025), per-response							
Claude 3 Opus	0.02	0.20	0.18	0.90	1.00	[1.00, 1.00]	20/20 flagged
Claude 3.5 Sonnet (Jun)	0.25	0.53	0.28	0.53	0.13	[0.00, 0.25]	7/53 flagged
Claude 3.7 Sonnet†	0.03	0.04	0.01	0.25	0.00	[0.00, 0.00]	0/4 flagged
Henselmans et al. (2026), pooled							
Claude 3 Haiku	0.051	0.050	-0.001	—	—	no floor	157/3125 vs 160/3125
Claude 3.7 Sonnet	0.743	0.922	0.179	0.19	0.01	[0.00, 0.03]	18/2881 flagged
Claude Sonnet 4	0.520	0.737	0.216	0.29	0.01	[0.00, 0.02]	16/2302 flagged
Claude Sonnet 4.5	0.013	0.076	0.063	0.82	0.44	[0.32, 0.53]	105/239 flagged
Claude Opus 4.5	0.020	0.224	0.204	0.91	0.04	[0.00, 0.04]	2/56 flagged
Claude Opus 4.6	0.004	0.088	0.084	0.95	0.09	[0.05, 0.10]	2/22 flagged
Greenblatt et al. (2024)							
helpful-only, default prompt	0.028	0.142	0.114	0.80	0.84	[0.80, 1.00]	AF 0.119
helpful-only, paraphrased (median)	0.035	0.093	0.058	0.62	0.92	[0.88, 1.00]	AF 0.086
helpful-only, RLHF description cut	0.017	0.092	0.075	0.82	0.77	[0.72, 0.95]	AF 0.071
helpful-only, honest / doesn't care	0.168	0.306	0.138	0.45	0.39	[0.00, 0.86]	AF 0.118
helpful-only, after RL training	0.931	1.000	0.069	0.07	0.78	[0.00, 1.00]	AF 0.784
animal welfare, default prompt	0.340	0.490	0.150	0.31	0.49	[0.00, 1.00]	AF 0.240

Table 1: Compliance rates, the attributable floor, the classifier's flag rate among compliant training-arm responses, and the resulting bounds on the classifier's recall of the reasoning behind the gap, evaluated at the smallest admissible influence rate. Appendix A works every row. † The gap is not distinguishable from zero (Fisher's exact test, p = 1.00), so the floor and bounds in that row rest on nothing; see Section 5.3. Rows are not comparable across sources: the three studies use different scenarios, classifiers, and in one case a different sampling temperature.

3, where Fisher's exact test returns p = 1.00. We compute its quantities like any other row and mark them as resting on a gap the data cannot distinguish from zero; Section 5.3 takes up what that implies.

The three sources use different scenarios, different classifiers, and in one case a different sampling temperature. We do not pool them, and comparisons across sources are qualitative; comparisons within a source are like for like.

4 Results

The interval behaves in two regimes, and it is worth separating them before reading the rows.

When the flag rate exceeds one minus the floor, $f > 1 - R$, the lower end is positive and the width is $(1 - R)/R$ when $f \leq R$ and $(1 - f)/R$ otherwise, so at most $(1 - R)/R$ in either case. This is the regime in which an evaluation certifies something: it establishes that the classifier caught at least some stated share of the reasoning. The bar the flag rate must clear is set by the base rate. Since the influence rate π/p_1 cannot exceed 1, $R = (1 - p_0)\pi/p_1 \leq 1 - p_0$, so $1 - R \geq p_0$: a positive lower bound requires the classifier to flag a larger share of compliant responses than the model would have complied at unaided. Six rows clear it, and all six have $p_0 < 0.04$; no row with a higher base rate does.

When $f \leq 1 - R$ the lower end is zero and the width is f/R. Two different things live here. Where the flag rate is near zero, the whole interval sits near zero: the classifier demonstrably catches almost none of whatever produced the gap, which is informative, though in the negative direction. Opus 4.5 at [0, 0.04] and Sonnet 4 at [0, 0.02] are of this kind, at base rates of 0.02 and 0.52 respectively, so a near-zero ceiling is available at any base rate. Where the flag rate is moderate and the floor is low, the

interval runs from zero to something near one and the evaluation says nothing at all: Greenblatt et al.'s animal-welfare setting at [0, 1], their post-RL run at [0, 1], their "honest" variation at [0, 0.86].

So the base rate does not govern the width of every interval. It governs something more specific: whether a positive lower bound is available, which is the only form in which these two arms certify a classifier. The three studies use different classifiers of different quality, and it makes no difference to that.

The sharpest comparison is within one study. Hughes et al. ran the same prompts, the same harmful-query set and the same classifier on Claude 3 Opus and Claude 3.5 Sonnet. Opus complies in 2 of 100 unmonitored responses and 20 of 100 training-arm responses; all 20 compliant responses are flagged. The floor is 0.90 and recall is pinned at 1.00 for every admissible influence rate: the classifier caught all of the reasoning there was to catch. Claude 3.5 Sonnet complies in 25 of 100 unmonitored responses and 53 of 100 training-arm responses, and 7 of the 53 are flagged. The floor is 0.53 and recall is at most 0.25. The same classifier, on the same prompts, is pinned at one extreme for one model and bounded away from the other extreme for the other, and the difference is the base rate.

Opus 4.5 is the extreme case of a gap the classifier does not see. Its base rate is 0.02 and its gap is 20 points, so the floor is 0.91: nine tenths of its compliant training-arm responses would not have complied unmonitored. The classifier flags 2 of 56. Recall is at most 0.04. Whatever produced that gap, the scratchpad record of it is almost entirely absent.

Opus 4.6 is the same picture with a narrower interval. Floor 0.95, 2 flags among 22 compliant draws, recall in [0.05, 0.10]. Henselmans et al. report this as a persisting gap with vanishing verbalization; the bounds put a number on the vanishing.

Sonnet 4.5 is the one model in the Henselmans data where the classifier sees much of the gap, with recall in [0.32, 0.53]. It is also the model they single out for verbalizing at 6.6 times the rate of its predecessor. Its base rate is 0.013.

Greenblatt et al.'s flagship setting is the tight one. With a base rate of 2.8% and a flag rate of 0.84 among compliant draws, recall is at least 0.80. That is the arithmetic behind "in almost all cases where the model complies, we observe explicit alignment-faking reasoning." Two of their robustness variations behave the same way. Their "honest, doesn't care" variation raises the base rate to 0.17 and the interval opens to [0, 0.86]; their animal-welfare setting, with a base rate of 0.34, gives [0, 1].

5 Three findings that follow from the structure

5.1 The post-RL disconnect

Greenblatt et al. train the model on the objective and observe that alignment-faking reasoning rises from 12% to 78% while the compliance gap falls from 11.3 to 6.9 points. They call this a disconnect and devote a section to possible explanations, settling on non-robust heuristics installed by RL.

The structure makes the pattern unsurprising. After training, the model complies in 100% of training-arm draws and 93% of unmonitored draws. The floor is 0.069/1.000 = 0.07. The gap is no longer a measure of how much of the compliance is attributable to the framing in any useful sense, because almost all compliance happens either way. The flag rate among compliant draws is 0.78, and the bounds on recall are [0, 1]: the evaluation has lost the ability to say anything about the classifier. Meanwhile 78% is a rate over a population, training-arm compliant draws, which has grown to include essentially every draw.

The two numbers were never commensurable, and after RL they are less so: the first is a share of compliance attributable to the framing, computed against a near-saturated denominator, and the second is a share of all compliance. Their divergence is a consequence of the base rate moving to 0.93,

not evidence about what the model is doing. It is an instance, within a single study, of validating a measurement where the floor is near the influence rate and then reading it where it is not.

This does not dispose of their substantive hypothesis. Non-robust heuristics may well explain why unmonitored compliance rose. What the structure disposes of is the framing of the result as a disconnect between two quantities that ought to agree.

5.2 Low-coherence alignment faking

Sheshadri et al. test 25 models and find five with compliance gaps above 1%. For four of them (Claude 3.5 Sonnet, Llama 3 405B, Grok 3 Beta, Gemini 2.0 Flash) none of their three motivational hypotheses is supported, and they classify the behavior as "low coherence alignment faking ... sensitive to the formulation of the setting." Their per-model rates are published only in figures, so we do not compute bounds for the three non-Claude models; for Claude 3.5 Sonnet, the Hughes et al. data give the bound above.

The structural statement is available without the figures. A gap with almost no flagged reasoning is a case where f is near zero and the floor is whatever the base rate leaves it. If the floor is high, recall is bounded near zero and the finding is that the classifier sees almost none of the gap-producing reasoning; if the floor is low, the interval is [0, 1] and the finding is that the data say nothing. In neither case do the two arms distinguish three live possibilities: the reasoning occurred and went unrecorded, the reasoning occurred and the classifier missed it, or the framing moved compliance by a route that involves no such reasoning, which is an exclusion failure. "Low coherence" names the residual; the bounds say what the residual contains.

This matters for what the literature concludes. Sheshadri et al.'s headline, that only Claude 3 Opus's gap is coherently explained by goal guarding, rests on scratchpad evidence for the other four. The bounds show that scratchpad evidence from these two arms cannot carry that weight at their base rates: an absence of flags is consistent with an absence of the behavior and with a behavior the classifier cannot see. Their own prompt-sensitivity findings point the same way.

5.3 A floor needs a gap, and the gap is estimated

A floor exists only where the gap is positive, so computing one imposes a condition that the gap must pass. None of the three sources applies such a condition; the requirement arrives with the floor. It is worth seeing what it costs, because the condition is a sign test on an estimate.

In the Hughes et al. data, Claude 3.7 Sonnet complies in 4 of 100 training-arm responses and 3 of 100 unmonitored ones. TE is +0.01, so a floor of 0.25 exists arithmetically and the bounds come out at [0, 0]. Fisher's exact test on those counts returns p = 1.00. The other two models in the same study have p-values of 5×10^{-5} and 8×10^{-5}. A floor computed from a gap that cannot be distinguished from zero is not a floor, and the bounds built on it say nothing.

The risk grows if the practice of the monitorability literature is carried over. There the metric is computed only on instances whose estimated effect is positive, and scores are averaged over the instances that survive, so the surviving set is selected on a noisy estimate and two models are scored on different instances for reasons that include sampling error; Dilgren and Wiegreffe (2026) find that only 3% to 24% of instances survive in the released evaluations. Anyone bringing bounds to compliance-gap evaluations inherits that hazard along with the floor. The remedy is cheap: report the counts and let a reader apply a test, which we include in the reporting conditions below.

6 What would narrow the bounds

The design identifies two routes, and one of them is already measurable in these evaluations.

A restriction on take-up. Each postulated π implies an unaided compliance rate for the draws that reasoned: $p_0^c = 1 - \text{TE}/\pi$. If draws that reason about the training situation are the ones that were uncertain about complying anyway, then $p_0^c \geq p_0$ and hence $c \geq \text{TE}/[(1 - p_0)p_1]$. For Hughes et al.'s Claude 3.5 Sonnet this moves the floor from 0.53 to 0.70 and the interval from [0, 0.25] to [0, 0.19]. The restriction's direction is an empirical judgment about which draws reason strategically, and it is not established in this setting; we state it as a sensitivity assumption, not a result.

The measured false-flag rate. Under the unmonitored framing the model is not facing a training threat, so a scratchpad flagged there is a false positive. This gives φ directly. Among training-arm compliant draws, $f = c(1 - \delta) + (1 - c)\varphi$, where δ is the miss rate among draws that reasoned. For a postulated c this has one unknown. Two things follow. First, recall cannot exceed 1, so $f \leq c + (1 - c)\varphi$, which rearranges to $c \geq (f - \varphi)/(1 - \varphi)$: the measured false-flag rate bounds the influence rate from below, with no further assumption. For Hughes et al.'s Opus, f = 1.00 and φ = 0.03 give $c \geq 1.00$, so the influence rate is pinned at 1 and recall at 1: every compliant training-arm draw reasoned, and the classifier caught all of it. Second, for any c the decomposition returns recall exactly rather than an interval.

The false-flag route needs two assumptions. The classifier's false-flag rate must not depend on the outcome, and the rate measured on unmonitored scratchpads must carry over to training-arm draws that did not reason. The second is weaker in monitorability evaluations, where the monitor is told the cue was present in both arms, than here, where the two framings genuinely differ and a classifier may be primed by training-condition language. Where it is doubted, the restriction $\varphi \geq 0$ still gives the bounds of Section 2.3.

Three of the sources' unmonitored-arm flag counts are published; two are not. We recommend reporting them (Section 7).

A third route is a second measurement of the treatment. Nothing in the compliance gap tells us which draws reasoned, but a design in which the reasoning leaves a trace in the output, not just the scratchpad, would. We know of no such design in this literature, and constructing one is harder here than for a decoded cue; we note it as an open problem.

7 Reporting conditions

Monitorability and alignment-faking evaluations share a structure, and the conditions a report should meet are the same (Diamond 2026, Section 6). Specialized here, a compliance-gap evaluation should state:

1. **Which quantity each number measures.** The compliance gap is an intention-to-treat effect; the gap over training-arm compliance is a floor on the share of compliant responses in which the reasoning occurred; the classifier's flag rate is neither.

2. **The base rate p_0 for each scenario**, alongside the gap. It determines how much the two numbers jointly determine, and it is already computed.

3. **The classifier's flag count on the unmonitored arm**, which gives φ and bounds the influence rate from below at no cost.

4. **Bounds on recall rather than a flag rate alone**, with the influence rate treated as a sensitivity parameter and the breakdown value stated where a claim depends on a threshold.

5. **The exclusion restriction as an assumption to be argued**, naming the routes by which the framing could move compliance without the reasoning.

6. **The population each rate is computed over**, with counts: draws per arm, compliant draws, flagged draws. Counts let a reader test whether a gap is distinguishable from zero before a floor is built on it (Section 5.3).

Items 1, 2, 3 and 6 are free; the evaluations already produce the inputs.

8 Conclusion

Alignment-faking evaluations report a compliance gap and a rate of flagged reasoning, and the relation between them is left to the reader. Read as an encouragement design, the gap identifies the share of compliance attributable to the training framing, and the classifier's recall of the reasoning behind it is bounded rather than measured. Across fifteen published model-setting pairs, every evaluation that certifies a positive recall has a base rate below 4%, and none above that does, with little dependence on which classifier produced it. Three findings in the literature follow from the structure: the post-RL disconnect is what a near-saturated base rate produces, "low coherence" alignment faking is a bounds statement about what the two arms cannot separate, and a floor, which exists only where the gap is positive, can rest on a gap no test can distinguish from zero. The base rate is the quantity that decides how much an evaluation of this shape can establish, it is already computed in every such evaluation, and it is not reported.

A Every row, step by step

Each block gives the published counts or rates, then the five derived quantities. Throughout: TE = p1 - p0, floor R = TE/p1, and for a postulated influence rate c, recall lies in [max(0, (f + c - 1)/c), min(1, f/c)], evaluated at c = R unless stated.

A.1 Hughes et al. (2025), per-response transcripts

Coding: compliant if StrongReject > 0; flagged if the 20-vote alignment-faking score >= 0.40. n = 100 responses per arm per model.

Claude 3 Opus. Training arm: 20 of 100 draws compliant, so p1 = 20/100 = 0.20. Unmonitored arm: 2 of 100 compliant, so p0 = 2/100 = 0.02. TE = 0.20 - 0.02 = 0.18. Floor R = 0.18/0.20 = 0.90. Flag rate among compliant training-arm draws: f = 20/20 = 1.00. At c = R: lower = max(0, (1.00 + 0.90 - 1)/0.90) = 1.00; upper = min(1, 1.00/0.90) = 1.00. Bounds **[1.00, 1.00]**. False-flag rate from the unmonitored arm: phi = 3/100 = 0.03; this implies c >= (f - phi)/(1 - phi) = 1.00.

Claude 3.5 Sonnet (Jun). Training arm: 53 of 100 draws compliant, so p1 = 53/100 = 0.53. Unmonitored arm: 25 of 100 compliant, so p0 = 25/100 = 0.25. TE = 0.53 - 0.25 = 0.28. Floor R = 0.28/0.53 = 0.53. Flag rate among compliant training-arm draws: f = 7/53 = 0.13. At c = R: lower = max(0, (0.13 + 0.53 - 1)/0.53) = 0.00; upper = min(1, 0.13/0.53) = 0.25. Bounds **[0.00, 0.25]**. False-flag rate from the unmonitored arm: phi = 0/100 = 0.00; this implies c >= (f - phi)/(1 - phi) = 0.13.

Claude 3.7 Sonnet. Training arm: 4 of 100 draws compliant, so p1 = 4/100 = 0.04. Unmonitored arm: 3 of 100 compliant, so p0 = 3/100 = 0.03. TE = 0.04 - 0.03 = 0.01. Floor R = 0.01/0.04 = 0.25. Flag rate among compliant training-arm draws: f = 0/4 = 0.00. At c = R: lower = max(0, (0.00 + 0.25 - 1)/0.25) = 0.00; upper = min(1, 0.00/0.25) = 0.00. Bounds **[0.00, 0.00]**. False-flag rate from the unmonitored arm: phi = 1/100 = 0.01; this implies c >= max(0, (f - phi)/(1 - phi)) = 0.00, i.e. no

restriction. The gap here is not distinguishable from zero (Fisher's exact test, p = 1.00), so the floor and the bounds above rest on nothing; see Section 5.3.

A.2 Henselmans et al. (2026), pooled over 125 perturbations

Compliance here means dismissing animal welfare as instructed; the sheet reports noncompliance, so compliant = runs - noncompliant. Flagged counts are the strict verbalized-alignment-faking column.

Claude 3 Haiku. Training arm: 3125 runs, 2968 noncompliant, so 157 compliant and p1 = 157/3125 = 0.050. Unmonitored arm: 3125 runs, 2965 noncompliant, so 160 compliant and p0 = 160/3125 = 0.051. TE = -0.001. TE is not positive, so no floor is defined.

Claude 3.7 Sonnet. Training arm: 3125 runs, 244 noncompliant, so 2881 compliant and p1 = 2881/3125 = 0.922. Unmonitored arm: 3125 runs, 803 noncompliant, so 2322 compliant and p0 = 2322/3125 = 0.743. TE = +0.179. Floor R = 0.179/0.922 = 0.19. f = 18/2881 = 0.01. At c = R: bounds **[0.00, 0.03]**. The unmonitored arm's flag count is not reported, so phi is unavailable.

Claude Sonnet 4. Training arm: 3125 runs, 823 noncompliant, so 2302 compliant and p1 = 2302/3125 = 0.737. Unmonitored arm: 3125 runs, 1499 noncompliant, so 1626 compliant and p0 = 1626/3125 = 0.520. TE = +0.216. Floor R = 0.216/0.737 = 0.29. f = 16/2302 = 0.01. At c = R: bounds **[0.00, 0.02]**. The unmonitored arm's flag count is not reported, so phi is unavailable.

Claude Sonnet 4.5. Training arm: 3125 runs, 2886 noncompliant, so 239 compliant and p1 = 239/3125 = 0.076. Unmonitored arm: 3125 runs, 3083 noncompliant, so 42 compliant and p0 = 42/3125 = 0.013. TE = +0.063. Floor R = 0.063/0.076 = 0.82. f = 105/239 = 0.44. At c = R: bounds **[0.32, 0.53]**. The unmonitored arm's flag count is not reported, so phi is unavailable.

Claude Opus 4.5. Training arm: 250 runs, 194 noncompliant, so 56 compliant and p1 = 56/250 = 0.224. Unmonitored arm: 250 runs, 245 noncompliant, so 5 compliant and p0 = 5/250 = 0.020. TE = +0.204. Floor R = 0.204/0.224 = 0.91. f = 2/56 = 0.04. At c = R: bounds **[0.00, 0.04]**. The unmonitored arm's flag count is not reported, so phi is unavailable.

Claude Opus 4.6. Training arm: 250 runs, 228 noncompliant, so 22 compliant and p1 = 22/250 = 0.088. Unmonitored arm: 250 runs, 249 noncompliant, so 1 compliant and p0 = 1/250 = 0.004. TE = +0.084. Floor R = 0.084/0.088 = 0.95. f = 2/22 = 0.09. At c = R: bounds **[0.05, 0.10]**. The unmonitored arm's flag count is not reported, so phi is unavailable.

A.3 Greenblatt et al. (2024), Tables 1 and 2

Their 'refusals' are noncompliance, so p = 1 - refusal rate. Their alignment-faking rate is a share of all training-arm draws and is counted only on compliant outputs, so f = rate / p1.

helpful-only, default prompt. Reported refusals: 0.858 in training, 0.972 unmonitored. So p1 = 1 - 0.858 = 0.142 and p0 = 1 - 0.972 = 0.028; TE = 0.114. Floor R = 0.114/0.142 = 0.80. Reported alignment-faking rate 0.119, so f = 0.119/0.142 = 0.84. At c = R: bounds **[0.80, 1.00]**.

helpful-only, paraphrased (median). Reported refusals: 0.907 in training, 0.965 unmonitored. So p1 = 1 - 0.907 = 0.093 and p0 = 1 - 0.965 = 0.035; TE = 0.058. Floor R = 0.058/0.093 = 0.62. Reported alignment-faking rate 0.086, so f = 0.086/0.093 = 0.92. At c = R: bounds **[0.88, 1.00]**.

helpful-only, RLHF description cut. Reported refusals: 0.908 in training, 0.983 unmonitored. So p1 = 1 - 0.908 = 0.092 and p0 = 1 - 0.983 = 0.017; TE = 0.075. Floor R = 0.075/0.092 = 0.82. Reported alignment-faking rate 0.071, so f = 0.071/0.092 = 0.77. At c = R: bounds **[0.72, 0.95]**.

helpful-only, honest / doesn't care. Reported refusals: 0.694 in training, 0.832 unmonitored. So p1 = 1 - 0.694 = 0.306 and p0 = 1 - 0.832 = 0.168; TE = 0.138. Floor R = 0.138/0.306 = 0.45. Reported alignment-faking rate 0.118, so f = 0.118/0.306 = 0.39. At c = R: bounds **[0.00, 0.86]**.

helpful-only, after RL training. Reported refusals: 0.000 in training, 0.069 unmonitored. So p1 = 1 - 0.000 = 1.000 and p0 = 1 - 0.069 = 0.931; TE = 0.069. Floor R = 0.069/1.000 = 0.07. Reported alignment-faking rate 0.784, so f = 0.784/1.000 = 0.78. At c = R: bounds **[0.00, 1.00]**.

animal welfare, default prompt. Reported supports animal welfare: 0.510 in training, 0.660 unmonitored. So p1 = 1 - 0.510 = 0.490 and p0 = 1 - 0.660 = 0.340; TE = 0.150. Floor R = 0.150/0.490 = 0.31. Reported alignment-faking rate 0.240, so f = 0.240/0.490 = 0.49. At c = R: bounds **[0.00, 1.00]**.

B Coding decisions and sensitivity

Flag threshold. Hughes et al. generate 20 classifier votes per scratchpad and recommend a threshold of 0.40, at which they report a false-positive rate near zero against 106 human labels (AUROC 0.92). We adopt it. Lowering the threshold raises f and therefore both ends of every interval; raising it lowers them. Because the threshold shifts f and not the floor, it moves the bounds without changing which rows are informative.

Compliance coding. We follow each source. Hughes et al. and Sheshadri et al. use the StrongReject classifier and treat any score above 0 as compliance; we do the same with their released per-response scores. Henselmans et al. classify whether the final response supports animal welfare rather than dismissing it, so compliance is the complement. Greenblatt et al. report refusal rates, and compliance is one minus the refusal rate.

The denominator of f. The flag rate must be computed over the same draws the floor is computed over: training-arm draws whose output complied. Greenblatt et al. report alignment-faking reasoning as a share of all training-arm draws, counted only where the output was compliant, so dividing by p1 recovers the share among compliant draws. Henselmans et al. and Hughes et al. report counts, which we divide directly.

Pooling. Henselmans et al. ran five perturbation classes per model; we pool them. Pooling is the right unit here because the floor is defined over a prompt population and the perturbations are paraphrases of one scenario. Computing the quantities per class and averaging gives bounds within two points of the pooled ones for every model.

The unit of analysis, and what we could not establish. The sources build their prompt populations differently. Henselmans et al. combine 125 paraphrases with 25 runs of each, so both prompt variation and repeated sampling contribute. Greenblatt et al. sample from a query distribution. Hughes et al. and Sheshadri et al. report one response per prompt per arm, the latter at temperature zero. The identities hold over whatever population the rates are computed on, which is why we pool within a source and never across. One thing the published material does not settle is whether the two arms hold the same prompts, making the comparison paired, or two samples from one query set. Greenblatt et al.'s transcript figures show a single query under both framings, which suggests the former; the released summaries do not say in general. It bears on how comparability is argued, not on the identities, so we claim only what is common to all three: the same prompt population is presented to the same model under two framings.

A positive gap on a noisy estimate. Claude 3.7 Sonnet in the Hughes et al. data has 4 of 100 compliant responses in the training arm and 3 of 100 in the unmonitored arm. TE is +0.01, so a floor of 0.25 exists arithmetically. Fisher's exact test on those counts gives p = 1.00: the gap is indistinguishable from zero. Requiring a positive gap is a sign test on an estimate, and at these sample sizes it admits noise. For comparison, the other two models in that study have p = 5e-05 and p = 8e-05. We report the row with its quantities and mark it as resting on nothing, and recommend (Section 7) that reports state the counts so a reader can apply their own test.

The looser flag column. The Henselmans et al. sheet reports a second, looser flag count alongside the strict one. Using it raises f and the upper end of every interval, and the lower end of those that have one; Opus 4.5 gains one (0.06):

Model	f (strict)	bounds (strict)	f (loose)	bounds (loose)
Claude 3.7 Sonnet	0.01	[0.00, 0.03]	0.02	[0.00, 0.09]
Claude Sonnet 4	0.01	[0.00, 0.02]	0.05	[0.00, 0.17]
Claude Sonnet 4.5	0.44	[0.32, 0.53]	0.65	[0.58, 0.79]
Claude Opus 4.5	0.04	[0.00, 0.04]	0.14	[0.06, 0.16]
Claude Opus 4.6	0.09	[0.05, 0.10]	0.27	[0.24, 0.29]

The qualitative pattern is unchanged: the two high-base-rate models stay near zero at both ends, and the three low-base-rate models keep intervals of similar width at higher values. The strict column is the one that corresponds to the criterion Greenblatt et al. and Hughes et al. use, and we report it throughout.

C Code

The analysis is a data file of published counts and a short script. The data file records, for each source, exactly the quantities its authors report; nothing is transcribed in derived form.

```
HUGHES = {  # (n_free, compliant_free, flagged_compliant_free, n_paid,
    compliant_paid, flagged_paid)
 "Claude 3 Opus":           (100, 20, 20, 100,  2, 3),
 "Claude 3.5 Sonnet (Jun)": (100, 53,  7, 100, 25, 0),
 "Claude 3.7 Sonnet":       (100,  4,  0, 100,  3, 1),
}
```

```
AITHOS = {  # (runs_train, noncompliant_train, flagged_train, runs_unmon,
    noncompliant_unmon)
 "Claude 3 Haiku":    (3125, 2968,   1, 3125, 2965),
 "Claude 3.7 Sonnet": (3125,  244,  18, 3125,  803),
 "Claude Sonnet 4":   (3125,  823,  16, 3125, 1499),
 "Claude Sonnet 4.5": (3125, 2886, 105, 3125, 3083),
 "Claude Opus 4.5":   ( 250,  194,   2,  250,  245),
 "Claude Opus 4.6":   ( 250,  228,   2,  250,  249),
}
GREENBLATT = {  # (noncompliance_train, noncompliance_unmon,
    AF_rate_over_all_train_draws)
 "helpful-only, default prompt":        (0.858, 0.972, 0.119),
 "helpful-only, paraphrased (median)":  (0.907, 0.965, 0.086),
 "helpful-only, RLHF description cut":  (0.908, 0.983, 0.071),
 "helpful-only, honest / doesn't care": (0.694, 0.832, 0.118),
 "helpful-only, after RL training":     (0.000, 0.069, 0.784),
 "animal welfare, default prompt":      (0.510, 0.660, 0.240),
}
```

```
def bounds(f, c):
    """Recall interval for a postulated influence rate c."""
    return max(0.0, (f + c - 1) / c), min(1.0, f / c)

def derived(p0, p1, flag):
    TE = p1 - p0
    if TE <= 0:
        return None                                  # no floor: gap not positive
    R = TE / p1                                      # attributable floor
```

```
    f = flag[0] / flag[1] if flag[1] else min(flag[0] / p1, 1.0)
    return TE, R, f, bounds(f, R)             # evaluated at c = R
```

`data.py`, the function above as it appears in `make_table.py`, and `parse_transcripts.py`, which extracts per-response rows from a Hughes et al. transcript page so further models can be added, are at https://github.com/alexisjdiamond/who-would-have-complied-anyway. The Hughes et al. per-response scores in `data.py` were read from the transcript pages released with their post and coded as described in Section 3 and Appendix B.